Session-XII: PHM data analysis: January last week or February 1st week

Program schedule

S.No	Time	Content	Method	Materials	Expected outcomes
1	15 min.	Recap.	Large group discussion	Eleventh session contents, charts and markers.	Participants recollect the contents discussed in eleventh session
2	45 min.	LTE data collection and analysis of results.	Large group discussion	FWS group kit, charts and markers.	Participants observe the changes in experimental and control plots and analyze the data.
3	20 min.	Data collection & analysis.	Small group & Large group discussion.	Charts and markers.	Participants analyze PHM data.
4	45 min.	HESA- Perceived change.	Small group discussion	Charts and markers.	Participants analyze the perceived change using HESA format
5	10 min.	FWS impact and success indicators.	Large group discussion	Charts and markers	Participants identify change or impact of FWS and list success indicators
6	15 min.	Group dynamics- PHM assets maintenance.	Story	Script of the story- The guru and his disciples	Participants understand the importance of maintaining PHM assets
7	10 min.	Homework: Actual rainfall received(Nov-May)	Large group discussion	Charts and markers	Participants collect actual rainfall received for Nov-May.
8	10 min.	Evaluation.	Large group discussion	Charts and markers.	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session participants will be able to:

- Recap the previous session,
- Observe LTE and collect data,
- Discuss and analyze PHM data,
- Identify perceived change and analyze using HESA,
- Discuss importance of PHM assets maintenance.

1. Recap of the previous session Time: 15 min.

Objective : Participants recap the contents discussed in the previous session.

Content : Contents of session eleven

- Observation of LTE and analysis,

- Soil moisture retention methods,
- Composting methods.
- Importance of fund mobilization.

Method : Small group discussion

Materials : Eleventh session contents, charts, and markers.

Process

- Before the start of the session, display visuals used in the 11th FWS session at appropriate place to help farmers in recap.
- Welcome participants to the 12th session and request participants to start the session with FWS pledge.
- After the FWS pledge, in the large group, encourage each participant to share a content that was discussed in 11th session and the learning from that one.
- As you facilitate the recap, write participants responses on the chart paper.
- Discuss briefly the contents and the learning from the previous session.

2. **LTE data collection and analysis of results** Time: 45 min.

Objective : Participants analyze the results of LTE.

Content : Results of LTE.

Method : Large group discussion

Materials : Charts and markers.

Process

- Start the discussion stating that we have successfully completed our Long Term Experiment. It is time for us to analyze the results of the experiment.
- Encourage the collaborator to share the practices followed in the experiment.
- Encourage the participants to discuss the differences between experimental plot and farmer practice plot.
- Encourage the participants to list out the factors that influenced the results.
- Analyze the factors that influenced the results.

3. **Data collection and analysis** Time: 20 min.

Objective : Participants discuss home work and analyze the PHM data.

Content :

- PHM data.
    - Number of functional wells.
    - Average water levels, pumping hours, pumping days and discharge.
- Analysis.

Method : Small group discussion and large group discussion

Materials : charts and markers.

Process

- Remind participants about the home work with regard to collection of PHM data.
- Request participants to work in small groups to discuss and list the reasons for increase or decrease in the PHM data on:
    i. Number of functional borewells.

ii. Average water levels and discharge of the borewells.
iii. Pumping hours per day and average pumping days.

- Request each group to present their group work and reasons for increase or decrease in the said data.
- Next, in large group consolidate the data for entire HU. And share the basic data that was collected before the start of FWS sessions.

4. HESA-Perceived change Time: 45 min.

- Encourage participants to list the reasons for fluctuations in water levels and discharge of the borewells. This will help facilitators understand the impact of farmer water schools and the concrete reasons for change.
- Use HESA chart for analysis. Discuss in detail. For ex: Number of bore wells increased. List the reasons for the increase and discuss whether it is beneficial or non beneficial. And why it is beneficial/non beneficial. List all the analysis on HESA chart and encourage participants to take decisions to reduce the loss.

5. **FWS Impact and success indicators** Time: 10 min.

Objective : Participants discuss the impact of FWS and identify success indicators.
Content : FWS Impact and success indicators.
Method : Large group discussion
Materials : Script for the story, charts and markers.
Process

- Ask participants the following question in large group. "What is your opinion on Farmer Water School and its approach? Do you feel FWS approach is useful in increase of knowledge and skill?"
- List participant's responses on a chart paper.
- If participants share that FWS approach is useful and successful, encourage them to identify changes in terms of quality and quantity.
- Share that these indicators will help to share with larger audience during field days and other public forums on the effectiveness of FWS approach in management of groundwater resources.

Tips to facilitators:

FWS impact in:

- Monitoring PHM data by HUNs and GMCs.
- Organizing Crop Water Budget exercise at HU level by HUNs.
- Decision making in GMCs and HUNs around groundwater and agriculture.
- Number of women facilitators in a HUN.
- Number of men and women farmer who are able to measure water levels and rainfall.

6. Group dynamic-PHM assets maintenance Time: 15 min.

Objective : Participants understand the importance of maintaining PHM assets.
Content : Importance of PHM assets maintenance.

Method : Story

Materials : Script for the story, charts and markers.

Process

- Long ago, there lived a guru who had three disciples.
- Once day, the guru decided to go on a pilgrimage.
- The guru's only property was three sacks of paddy which he decided to give one each to his three disciples.
- While giving a sack of paddy to each of his disciples, the guru suggested them to use it in the best way possible with out wasting it and left on a pilgrimage.
- The first disciple assumed that his guru might take a long time to return from pilgrimage or not return at all. With this idea, he milled the paddy and ate up all the rice.
- The second disciple was worried that some animal might eat the paddy or a thief would steal it. So, he kept it beside him all the time and even slept on it. The paddy gradually deteriorated and was destroyed.
- The third disciple sowed the paddy. By the time the Guru returned from his pilgrimage, he was able to harvest two crops, producing 40 bags of paddy.
- Pose a question in large group: "Which disciple utilized the paddy in the best possible way?"
- List out the participants responses on a chart paper.
- Summarize the discussion by stating that 'any asset is useless if it is spent or kept idle. It will give us maximum benefit generating revenue only when utilized in a productive way'.

7. **Homework: Actual rainfall data received (Nov-May)** Time: 10 min.

- Discuss the need for actual rainfall received from Nov-May.
- Share that the data will help to estimate actual groundwater recharge for Nov-May and the change in estimated groundwater balance at the end of May during the CWB workshop.

8. **Evaluation** Time: 10 min.

Ask the following questions to evaluate the day's learning. Pose the questions to the large group:

- Why do we analyze the data?
- What are the methods used to analyze the data?